Chapter 9. Regression Discontinuity, Difference-in-Differences and Synthetic Control

Sharp and fuzzy RD, DiD and staggered adoption, synthetic control and synthetic DiD

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9.1 Regression discontinuity

Motivation. How does a known cutoff rule identify a causal effect at the cutoff, and how does the bandwidth drive bias, variance and inference?

9.1.1 Where RDD fits

Motivation. Place RDD among empirical designs.

Types of empirical studies, from more to less researcher control:

Experiment (RCT)	Natural experiment	Quasi-experiment	Observational study
researcher randomizes treatment	as-if random assignment by nature/policy (lottery, IV)	known non-random rule exploited (RD, DiD, SCM)	confounders must be assumed away (matching, IPW)

Method	Design type	Key source of variation
Back-door / matching / IPW	observational	adjusted confounders
IV	natural / quasi-experiment	exogenous instrument
RDD	quasi-experiment	cutoff rule
DiD	quasi-experiment	temporal variation across groups
SCM	quasi-experiment	weighted donor pool

IV straddles the boundary (lottery vs. policy threshold). Here treatment is determined by whether a pre-treatment variable crosses a known threshold.

Notation. $Z_i \in \mathbb{R}$: forcing (running) variable (not an instrument as in Chapter 7); c: cutoff; $X_i = \mathbb{1}\{Z_i \geq c\}$: treatment in the sharp design; $Y_i(1), Y_i(0)$: potential outcomes; τ_{SRD}: sharp-RD effect at the cutoff; $W_i = \mathbb{1}\{Z_i \geq c\}$: the instrument in the fuzzy design.

9.1.2 Concept and identification

Motivation. Define the sharp RD effect and show that continuity identifies it.

9.1.2.1 Examples

Incumbency advantage (Lee 2008; Lee et al. 2004). Running variable: Democratic vote-share margin at t; cutoff 50%; outcome: Democratic vote share at $t+1$. Near the cutoff the election outcome is as good as random; barely winning raises next-election vote share by 6–8 points.

Scholarships and test scores (Thistlethwaite and Campbell 1960, the first RDD): eligibility depends only on an exam score exceeding a threshold. **Financial aid**: eligibility if family income is below a threshold.

9.1.2.2 The design

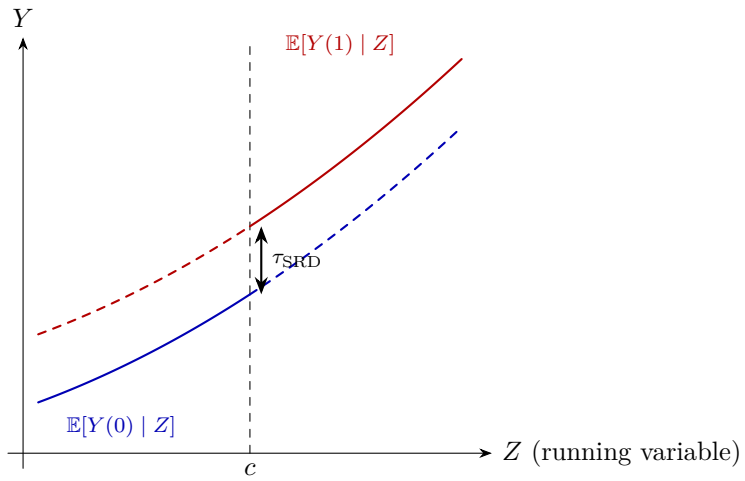


Figure. *Sharp RD:* $\mathbb{E}[Y(0) \mid Z]$ *is observed only left of* c, $\mathbb{E}[Y(1) \mid Z]$ *only right (solid). Under continuity the jump at* c *is the effect.*

Treatment changes discontinuously in a pre-treatment forcing variable while confounders vary smoothly; comparing units with similar Z_i but different treatment identifies a *local* effect at the cutoff.

Sharp RD. Z_i continuous (discrete running variables: Cattaneo et al. 2020) and $X_i = \mathbb{1}\{Z_i \geq c\}$ deterministic. Positivity fails by construction:

$$P[X_i = 1 \mid Z_i = c - \varepsilon] = 0, \qquad P[X_i = 1 \mid Z_i = c + \varepsilon] = 1,$$

so back-door adjustment and IPW fail; continuity-based extrapolation is used instead.

Fuzzy RD. The *probability* of treatment jumps at c, but not from 0 to 1 (Section 9.1.5).

9.1.2.3 Estimand and identification

$$\tau_{\text{SRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid Z_i = c] = \mathbb{E}[Y_i(1) \mid Z_i = c] - \mathbb{E}[Y_i(0) \mid Z_i = c].$$

No unit has $Z_i = c$ exactly, so local extrapolation needs an assumption.

Continuity assumption. $\mu_1(z) = \mathbb{E}[Y_i(1) \mid Z_i = z]$ and $\mu_0(z) = \mathbb{E}[Y_i(0) \mid Z_i = z]$ are continuous in z. Then

$$\begin{aligned}
\mathbb{E}[Y_i(0) \mid Z_i = c] &= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid Z_i = z] && \text{(continuity)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i(0) \mid X_i = 0, Z_i = z] && \text{(SRD: } z < c \Rightarrow X_i = 0\text{)} \\
&= \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z] && \text{(consistency),}
\end{aligned}$$

and likewise $\mathbb{E}[Y_i(1) \mid Z_i = c] = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$. Hence

$$\boxed{\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z].}$$

Estimation means two regression functions at a boundary point; nonparametric regression is consistent but slow.

What can go wrong? Sorting: richer students retake an exam more often to pass the aid threshold, so those just above differ from those just below.

9.1.3 Validity and diagnostics

Motivation. Check the design's assumptions graphically and with falsification tests.

Three standard figures:

1. **Outcomes by running variable:** a visible jump at c ("hard to sell" otherwise; Imbens and Lemieux 2008).
2. **Covariates by running variable:** should be smooth across c; a jump suggests invalid RD.
3. **Density of the running variable:** a discontinuity suggests sorting.

Manipulation and the McCrary test. If units can precisely manipulate Z_i, smoothness likely fails. McCrary (2008) tests for a jump of $f_Z(z)$ at c.

Bunching. Administrative data often show units locating just below thresholds (taxable earnings, full-time hours; World Bank blog "Ready, set, bunch"), e.g. reported wealth in Colombia bunching below a new wealth-tax threshold in 2010 (Londoño-Vélez).

Falsification (placebo) tests. A valid RDD shows an effect only at the true cutoff.

- *Placebo cutoffs:* re-estimate at fake cutoffs; in a simulation with an effect only at $c = 0$, placebo estimates on $[-2, 2]$ are near zero.
- *Placebo outcomes:* pre-treatment covariates as outcomes; a jump signals sorting.

9.1.4 Estimation

Motivation. Estimate boundary limits with local polynomial regression and choose the bandwidth.

9.1.4.1 General strategy

The target is a single *boundary* point with data on one side only, so boundary bias shrinks slowly; the naive difference in means uses data too far away. Write $\mu_+(c) = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z]$, $\mu_-(c) = \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$, $\tau_{\text{SRD}} = \mu_+(c) - \mu_-(c)$.

Kernel (local average) regression. With a uniform kernel and bandwidth h,

$$\hat{\mu}_-(c) = \frac{\sum_{i=1}^n Y_i \mathbb{1}\{c - h \le Z_i < c\}}{\sum_{i=1}^n \mathbb{1}\{c - h \le Z_i < c\}}$$

(similarly above), or equivalently

$$(\hat{\alpha}, \hat{\tau}_{\text{SRD}}) = \arg\min_{\alpha, \tau} \sum_{i: Z_i \in [c-h, c+h]} (Y_i - \alpha - \tau X_i)^2,$$

the difference in means, assuming locally constant outcomes. Large h: high bias, low variance; small h: the reverse; bias shrinks slowly. Kernels: uniform $K(u) = \frac{1}{2}\mathbb{1}\{|u| < 1\}$, triangular $K(u) = (1 - |u|)\mathbb{1}\{|u| < 1\}$.

9.1.4.2 Local linear regression

Allow a slope. On $Z_i \in [c - h, c)$,

$$(\hat{\alpha}_-, \hat{\beta}_-) = \arg\min_{\alpha, \beta} \sum_{i: Z_i \in [c-h, c)} \big(Y_i - \alpha - \beta(Z_i - c)\big)^2,$$

and the same on $[c, c+h]$. Then

$$\hat{\tau}_{\text{SRD}} = \hat{\mu}_+(c) - \hat{\mu}_-(c) = \hat{\alpha}_+ + \hat{\beta}_+(c-c) - \hat{\alpha}_- - \hat{\beta}_-(c-c) = \hat{\alpha}_+ - \hat{\alpha}_-.$$

Single regression. Equivalently,

$$\underset{(\alpha,\beta,\tau,\gamma)}{\arg\min} \sum_{i:Z_i\in[c-h,c+h]} \{Y_i - \alpha - \beta(Z_i-c) - \tau X_i - \gamma(Z_i-c)X_i\}^2,$$

with $\hat{\tau}_{\text{SRD}} = \hat{\tau}$: the fit is $\alpha + \beta(Z_i - c)$ for $X_i = 0$ and $(\alpha+\tau) + (\beta+\gamma)(Z_i - c)$ for $X_i = 1$.
Kernel weighting (typically triangular):

$$\underset{(\alpha,\beta,\tau,\gamma)}{\arg\min} \sum_{i=1}^{n} K\left(\frac{Z_i - c}{h}\right) \{Y_i - \alpha - \beta(Z_i-c) - \tau X_i - \gamma(Z_i-c)X_i\}^2.$$

9.1.4.3 Bias–variance trade-off and the MSE-optimal bandwidth

Let $\mu(z) = \mathbb{E}[Y \mid Z = z]$ with one-sided second derivatives $\mu_\pm^{(2)}(c)$, density $f(c)$, and $\sigma_\pm^2(c) = \text{Var}(Y \mid Z = c^\pm)$. For local linear regression,

$$\text{Bias} \approx C_1 h^2 \underbrace{(\mu_+^{(2)}(c) - \mu_-^{(2)}(c))}_{\text{curvature difference}}, \qquad \text{Variance} \approx \frac{C_2}{Nh} \cdot \underbrace{\frac{\sigma_+^2(c) + \sigma_-^2(c)}{f(c)}}_{\text{noise / density at } c}.$$

Minimizing MSE = Bias^2 + Variance (Imbens and Kalyanaraman 2012):

$$h_{\text{MSE}} = C_K \left(\frac{\sigma_+^2(c)/f_+(c) + \sigma_-^2(c)/f_-(c)}{\left(\mu_+^{(2)}(c) - \mu_-^{(2)}(c)\right)^2} \right)^{1/5} N^{-1/5},$$

since $\frac{d}{dh}[C_1^2 h^4 D^2 + C_2 V/(Nh)] = 0$ gives $h^5 \propto V/(ND^2)$. All inputs are estimated, curvature by local quadratic regression.

9.1.4.4 Why the MSE-optimal bandwidth may fail for inference

$h_{\text{MSE}} \propto N^{-1/5}$ is wide enough that bias is not negligible relative to the SE: a nominal 95% CI may cover about 80%. Remedies: *robust bias correction* (RBC; Calonico, Cattaneo and Titiunik 2014) subtracts an estimate of the leading bias and accounts for its variability; the *CE-optimal bandwidth* (Calonico, Cattaneo and Farrell 2020) minimizes coverage error, $h_{\text{RBC}} \propto N^{-1/4}$. Rule of thumb $\hat{h}_{\text{RBC}} \approx (0.7\text{–}0.8) \times \hat{h}_{\text{MSE}}$.

Bandwidth	Rate	Bias	Variance	CI length	Coverage
MSE-optimal	$N^{-1/5}$	high	low	short	fails
Undersmoothing	$N^{-1/3}$	very low	high	too long	valid
CE-optimal	$N^{-1/4}$	moderate	moderate	efficient	optimal

CE-optimal with RBC is the `rdrobust` default.

Bandwidth sensitivity. Credible estimates are stable across bandwidths (simulation with $\tau = 3$: small h gives wide CIs, large h bias).

9.1.4.5 Covariates and software

Covariates are not needed for identification but improve precision:

$$Y_i = \alpha + \beta(Z_i - c) + \tau X_i + \gamma(Z_i - c)X_i + \delta^\top V_i + \varepsilon_i,$$

with V_i pre-treatment (Calonico et al. 2019); a jump in V_i at c is evidence against validity.

`rdrobust` (R/Stata) implements MSE- and CE-optimal bandwidths, RBC CIs, covariates, clustering and fuzzy RD: `library(rdrobust); rd <- rdrobust(Y, Z, c = 1500); summary(rd)`.

9.1.5 Fuzzy RD

Motivation. Treat imperfect compliance with the cutoff as a local instrumental-variable problem.

Concept. Crossing the cutoff changes the probability of treatment,

$$\lim_{z \downarrow c} P[X_i = 1 \mid Z_i = z] \neq \lim_{z \uparrow c} P[X_i = 1 \mid Z_i = z],$$

but not from 0 to 1 (exemptions, imperfect compliance, administrative errors; e.g. not all eligible students attend summer school).

Fuzzy RD as IV: the Wald estimator. With instrument $W_i = \mathbb{1}\{Z_i \geq c\}$ for X_i, under continuity

$$\tau_{\text{FRD}} = \frac{\lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]}{\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z]} = \frac{\text{ITT effect on } Y}{\text{first-stage effect on } X}.$$

Assumptions. (1) Continuity of $\mathbb{E}[Y_i(x) \mid Z_i = z]$ and $\mathbb{E}[X_i(w) \mid Z_i = z]$ at c;

(2) first stage: $\lim_{z \downarrow c} \mathbb{E}[X_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[X_i \mid Z_i = z] \neq 0$;

(3) monotonicity: no defiers. Exclusion is built in (W_i affects Y_i only through X_i near c).

Interpretation. $\tau_{\text{FRD}} = \mathbb{E}[Y_i(1) - Y_i(0) \mid \text{complier}, Z_i = c]$: a LATE local to compliers *and* to the cutoff; strong internal, limited external validity.

9.1.6 Case study: remedial education in Chicago

Motivation. See a fuzzy RD in action (Jacob and Lefgren 2004, REStat).

Background. From 1996 Chicago Public Schools required minimum ITBS reading and math scores to advance (third graders: 2.8 grade equivalents); students below had to attend summer school and could be retained. *Question:* the effect of summer school and retention on achievement.

Design. Fuzzy (exemptions, discretion), but the treatment probability jumps sharply at the cutoff. Data: CPS records 1993–1999, about 350,000 third and sixth graders; outcomes: changes in ITBS scores (logits) one and two years later.

IV estimation. With $W_i = \mathbb{1}\{\text{Score}_i \geq \text{cutoff}\}$ (falling below triggers treatment, so the first-stage jump is negative),

$$\begin{aligned} \text{First stage:} \quad & \text{Treat}_i = \alpha_0 + \alpha_1 W_i + \alpha_2 \text{Score}_i + \alpha_3 (W_i \times \text{Score}_i) + \nu_i, \\ \text{Second stage:} \quad & Y_i = \beta_0 + \tau \widehat{\text{Treat}}_i + \beta_1 \text{Score}_i + \beta_2 (W_i \times \text{Score}_i) + \varepsilon_i, \end{aligned}$$

with τ the LATE near the cutoff; the interactions allow different slopes on each side.

Results. Third graders: about 0.11–0.13 logits in year one (about 20% of a year's learning), persisting with some fade-out; sixth graders: no reading effect, small insignificant math effect.

Retention alone: short-term gains in grade 3, negative reading effects in grade 6. Summer school alone: positive for third graders, especially low performers.

Strengths and limitations. A policy cutoff with fuzzy-RD IV and heterogeneity by grade, gender and SES; but LATE only, no 3+ year outcomes, and manual bandwidth choice.

9.1.7 Summary

- RDD exploits a known cutoff rule for treatment assignment.
- Identification: continuity at the cutoff, $\tau_{\text{SRD}} = \lim_{z \downarrow c} \mathbb{E}[Y_i \mid Z_i = z] - \lim_{z \uparrow c} \mathbb{E}[Y_i \mid Z_i = z]$.
- Diagnostics: outcome plot, covariate balance, McCrary test, placebo tests.
- Estimation: kernel-weighted local linear regression.
- Bandwidth: MSE-optimal ($N^{-1/5}$) for estimation, CE-optimal ($N^{-1/4}$) for inference; shrink h_{MSE} by 20–30%.
- Fuzzy RD = IV at the cutoff: LATE for compliers at c.
- Chicago: positive effects for third graders, little for sixth graders.

9.1.8 Appendix: Kink RD

Sharp kink RD looks for a jump in the slope of $\mathbb{E}[Y_i \mid Z_i = z]$ at c (Card, Lee, Pei and Weber 2015); local quadratic regression is preferable.

9.1.9 Appendix: Assumptions for the MSE-optimal bandwidth

(Imbens and Kalyanaraman 2012.) (1) $\{Y_i, Z_i\}$ i.i.d.;

(2) the density $f(z)$ is continuous with $f(c) > 0$;

(3) $\mu(z)$ is three times differentiable near c;

(4) K is symmetric, bounded, with compact support;

(5) $\sigma^2(z)$ is continuous near c with $\sigma^2(c) > 0$.

9.1.10 Appendix: Regularization of the bandwidth

If the estimated curvature difference is near zero, h explodes; IK (2012) add a penalty $\hat{r}$ (the variance of the curvature estimate):

$$\hat{h}_{\text{reg}} \propto \left(\frac{\text{sum of variances}}{f(c)\,[(\text{curvature})^2 + \text{regularization}]} \right)^{1/5},$$

with $\hat{\mu}_{\pm}^{(2)}(c)$ from local quadratic regressions and

$$\hat{r}_+ = \frac{2160\,\hat{\sigma}_+^2(c)}{N_{h,+}h^4}, \qquad \hat{r}_- = \frac{2160\,\hat{\sigma}_-^2(c)}{N_{h,-}h^4},$$

growing with less data or more noise.

9.1.11 Appendix: Why local linear? Polynomial order

Higher order p reduces bias but increases variance near c; in finite samples the variance cost dominates. Local linear ($p = 1$) gives the main estimate; local quadratic ($p = 2$) estimates the bias for the MSE-optimal bandwidth (IK 2012) and RBC (CCT 2014): "estimate with p, correct bias with $p + 1$" (`rdrobust` default).

9.1.12 Appendix: Global polynomials: a cautionary tale

Global high-order polynomials over the whole range (Gelman and Imbens 2019) are sensitive to far observations, oscillate (Runge-type behavior) and give poor CI coverage; local linear fits with a data-driven bandwidth avoid this. (Further reading: *The Effect*; *The Mixtape*.)

9.2 Difference-in-differences

Motivation. How do repeated measurements identify the ATT when the untreated trend is a valid proxy for the treated group's counterfactual trend?

9.2.1 Motivation

Motivation. Introduce the two sources of variation in panel data.

Panel data give within-unit variation over time and cross-sectional variation in treatment. Before–after uses the treated group's own past as counterfactual; DiD uses the control group's change as the treated group's counterfactual change (parallel paths).

9.2.1.1 Case study: minimum wages (Card and Krueger 1994)

In 1992 New Jersey raised its minimum wage from \$4.25 to \$5.05 while neighboring Pennsylvania stayed at \$4.25. Employment at fast-food restaurants in eastern PA and NJ (similar prices, wages, products) is observed before and after; the ATT is the gap between NJ's actual and counterfactual (parallel to PA) post-period levels.

9.2.2 The DiD design

Motivation. Formalize the two-group, two-period design and identify the ATT.

9.2.2.1 Setup

Two groups, two periods; only the group $G_i = 1$ is treated, at $t = 1$: $X_{i0} = 0$, $X_{i1} = G_i$.

	Pre-period ($t = 0$)	Post-period ($t = 1$)
Treated group ($G_i = 1$)	$X_{i0} = 0$	$X_{i1} = 1$
Control group ($G_i = 0$)	$X_{i0} = 0$	$X_{i1} = 0$

Potential outcomes. $Y_{it}(x)$; consistency $Y_{it} = X_{it}Y_{it}(1) + (1 - X_{it})Y_{it}(0)$. Hence $Y_{i0}(0) = Y_{i0}$ for all; $Y_{i1} = Y_{i1}(1)$ if $G_i = 1$; $Y_{i1} = Y_{i1}(0)$ if $G_i = 0$.

9.2.2.2 The identification problem

$$\text{ATT} = \mathbb{E}[Y_{i1}(1) - Y_{i1}(0) \mid G_i = 1] = \underbrace{\mathbb{E}[Y_{i1} \mid G_i = 1]}_{(a)} - \underbrace{\mathbb{E}[Y_{i1}(0) \mid G_i = 1]}_{(b)}.$$

(a) is estimable;

(b) is counterfactual.

9.2.2.3 Three control strategies

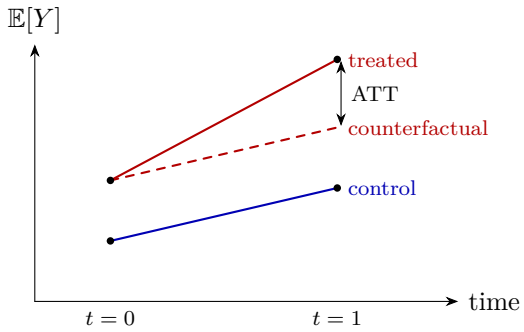


Figure. *DiD: absent treatment the treated group would change like the control group (dashed); the ATT is the difference of the two changes.*

1. **Before–after:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i0} \mid G_i = 1]$ (no change over time absent treatment).
2. **Cross-sectional comparison:** assume $\mathbb{E}[Y_{i1}(0) \mid G_i = 1] = \mathbb{E}[Y_{i1} \mid G_i = 0]$ (no group differences).
3. **Difference-in-differences:** assume parallel trends; then

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i0} \mid G = 1]) - (\mathbb{E}[Y_{i1} \mid G = 0] - \mathbb{E}[Y_{i0} \mid G = 0]).$$

9.2.2.4 Parallel trends

Definition (Parallel trends)

$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1]$.

It allows time-invariant unobserved group differences and common trends, but is *not invariant to nonlinear transformations*: parallel trends in Y_{it} imply non-parallel trends in $\log Y_{it}$, and vice versa.

Theorem 9.1 (Identification)

Under consistency and parallel trends,

$$\text{ATT} = (\mathbb{E}[Y_{i1} \mid G_i = 1] - \mathbb{E}[Y_{i0} \mid G_i = 1]) - (\mathbb{E}[Y_{i1} \mid G_i = 0] - \mathbb{E}[Y_{i0} \mid G_i = 0]).$$

Proof [Supplement]. By parallel trends, $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0}(0) \mid G = 1] + \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G = 0]$. Consistency gives $Y_{i0}(0) = Y_{i0}$ for everyone and $Y_{i1}(0) = Y_{i1}$ for controls, so $\mathbb{E}[Y_{i1}(0) \mid G = 1] = \mathbb{E}[Y_{i0} \mid G = 1] + \mathbb{E}[Y_{i1} - Y_{i0} \mid G = 0]$. Substitute into $\text{ATT} = \mathbb{E}[Y_{i1} \mid G = 1] - \mathbb{E}[Y_{i1}(0) \mid G = 1]$. ■

Threats: non-parallel trends (time-varying confounding). *Ashenfelter's dip:* earnings fall just before training-program entry, lowering the treated pre-period outcome and overestimating the effect. *Falsification test:* parallel pre-treatment trends; necessary, not sufficient.

9.2.2.5 Estimation

With panel data,

$$\widehat{\text{ATT}} = \underbrace{\frac{1}{N_1}\sum_{i=1}^{N} G_i\{Y_{i1} - Y_{i0}\}}_{\text{average trend, treated}} - \underbrace{\frac{1}{N_0}\sum_{i=1}^{N}(1-G_i)\{Y_{i1} - Y_{i0}\}}_{\text{average trend, control}},$$

with difference-in-means standard errors. Equivalently, $\Delta Y_i = Y_{i1} - Y_{i0} = \alpha + \tau G_i + \varepsilon_i$ gives $\hat{\tau}_{\text{OLS}} = \widehat{\text{ATT}}$, convenient for covariates. DiD also works with repeated cross-sections.

Card and Krueger with numbers (average full-time-equivalent employment per restaurant):

	Before (Feb 1992)	After (Nov 1992)	Difference
NJ (treated)	20.44	21.03	+0.59
PA (control)	23.33	21.17	−2.16
DiD estimate			+2.76

DiD $= (21.03 - 20.44) - (21.17 - 23.33) = 0.59 - (-2.16) = 2.76$: no evidence of employment loss.

9.2.2.6 DiD and two-way fixed effects (2×2)

$$Y_{it} = \mu + \gamma G_i + \delta t + \tau X_{it} + \varepsilon_{it}.$$

Since $X_{it} = G_i \cdot t$, the four cell means are μ, $\mu+\gamma$, $\mu+\delta$, $\mu+\gamma+\delta+\tau$, so $\hat{\tau}$ equals the DiD estimate.

9.2.3 Conditional and nonlinear DiD

Motivation. Weaken parallel trends with covariates, and replace mean trends by rank preservation.

9.2.3.1 Conditional parallel trends

For all x,

$$\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1, X_i = x] = \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0, X_i = x]:$$

trends need only be parallel within covariate strata. Estimators:

- **Outcome regression:** include X_i in the (TWFE) model.
- **IPW-DiD:** reweight controls by $\hat{e}(X_i)/(1-\hat{e}(X_i))$.
- **Doubly robust DiD:** consistent if either model is correct (Sant'Anna and Zhao 2020; Callaway and Sant'Anna 2021), as in Chapter 6.

9.2.3.2 Nonlinear DiD (changes-in-changes)

Assume parallel trends in *ranks*: absent treatment, a unit at the q-th quantile of its group in period 0 stays at the q-th quantile in period 1. This is invariant to monotone transformations.

Formalization (Athey and Imbens 2006). Let $F_{gt}(y) = P(Y_{it}(0) \le y \mid G_i = g)$. Observed: F_{00}, F_{01}, F_{10}, and the CDF $F_{11}^{(1)}$ of $Y_{i1}(1)$ for the treated. Goal: the counterfactual F_{11}. Quantile treatment effect:

$$\tau(q) = (F_{11}^{(1)})^{-1}(q) - F_{11}^{-1}(q).$$

Let $h_0(y) = F_{01}^{-1}(F_{00}(y))$ (controls) and $h_1(y) = F_{11}^{-1}(F_{10}(y))$ (treated). *Identification assumption:* $h_0 = h_1$. Then $F_{11}^{-1}(F_{10}(y)) = F_{01}^{-1}(F_{00}(y))$, and solving,

$$F_{11}(y) = F_{10}(F_{00}^{-1}(F_{01}(y))).$$

(Take $y' = F_{01}^{-1}(F_{00}(y''))$; the assumption says $F_{11}(y') = F_{10}(y'')$ with $y'' = F_{00}^{-1}(F_{01}(y'))$.)

9.2.3.3 Key takeaways

- Parallel trends are the core of DiD identification.
- DiD handles time-invariant confounding but not time-varying confounders (Ashenfelter's dip).
- Conditional parallel trends lead to IPW-DiD and doubly robust DiD.
- Pre-trend tests do not validate parallel trends; sensitivity analysis is preferred.
- Linear DiD is not invariant to nonlinear transformations; nonlinear DiD needs distributional assumptions.
- TWFE equals DiD in the 2×2 case but needs care with staggered timing.

9.2.4 Appendix: Pre-testing bias

Proceeding only after an insignificant pre-trend test biases the estimate: low-power tests select samples whose pre-trends happen to look flat, and that chance variation persists into the post-period. Under-rejection of the pre-trend test leads to over-rejection of "no effect" (Roth 2022).

9.2.5 Appendix: Sensitivity analysis for parallel trends

Allow bounded violations,

$$\left|\mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 1] - \mathbb{E}[Y_{i1}(0) - Y_{i0}(0) \mid G_i = 0]\right| \leq M,$$

build a confidence set for the ATT for each M, and report the *breakdown value*, the smallest M whose set includes 0. Bounds can be calibrated from pre-treatment trend differences (Rambachan and Roth 2023).

9.2.6 Appendix: Staggered treatment timing

With staggered entry, TWFE is a weighted average of all 2×2 DiDs (Goodman-Bacon): early vs. never-treated and late vs. never-treated (good), and late vs. early-treated (dangerous). If effects change over time, the early-treated are invalid controls, and the estimate is attenuated or even reversed.

Modern solutions. Callaway and Sant'Anna (2021): group-time ATTs with never- or not-yet-treated controls. Event-study plots by $t - t^*$ show pre-trends and dynamics. (See the lecture on staggered DiD.)

9.3 Staggered difference-in-differences

Motivation. How should DiD be done when adoption times differ and effects vary across units and over time, where TWFE can mislead?

9.3.1 Staggered adoption

Motivation. Describe the setting and why it complicates DiD.

Units are treated at different times (groups A, B, C, and a never-treated group); after adoption treated paths jump and may keep changing (dynamic effects); with treatment reversal effects switch on and off.

Staggered DiD handles gradual roll-outs: a *staggered adoption design* (SAD). Not-yet-treated units serve as controls, blurring treatment/control and pre/post. Once treated, a unit remains treated.

9.3.1.1 TWFE as is is not sufficient

TWFE $Y_{it} = \alpha_i + \beta_t + \tau D_{it} + \varepsilon_{it}$ recovers the ATT (under parallel trends and no anticipation) only if (i) effects are homogeneous or (ii) there are two periods. Otherwise already-treated units act as controls and their dynamics leak into τ. Fix: use only "good" comparisons.

9.3.1.2 Terminology

- **Static effects:** constant across post-treatment periods.
- **Dynamic effects:** change with time since adoption.
- **Homogeneous effects:** identical across units.
- **Heterogeneous effects:** vary across units, constant over time for each unit.
- **Carryover effects:** a past treatment keeps acting after withdrawal (a dynamic effect).

9.3.1.3 Valid and invalid comparisons

- **Valid:** treated vs. never-treated; treated vs. not-yet-treated.
- **Invalid:** treated vs. already-treated; later-treated vs. earlier-treated using the earlier group's post-treatment periods.

Bias arises under dynamic or heterogeneous effects (Goodman-Bacon decomposition, Section 9.2).

Modern alternatives to TWFE: Callaway and Sant'Anna (2021), group-time ATTs; Sun and Abraham (2021), event studies with proper controls; stacked DiD, one DiD per cohort.

9.3.2 Callaway–Sant'Anna group-time ATTs

Motivation. Define, identify and aggregate effects by adoption cohort and calendar time.

9.3.2.1 Notation

G_i: period in which i becomes treated; $Y_{it}(0)$: untreated potential outcome; $Y_{it}(g)$: potential outcome if first treated in g. Never-treated units have $Y_{it} = Y_{it}(0)$; for others

$$Y_{it} = \mathbb{1}\{G_i > t\} Y_{it}(0) + \mathbb{1}\{G_i \leq t\} Y_{it}(G_i),$$

which embeds no anticipation. C_i: never-treated indicator; D_{it}: treated by t; X_i: pre-treatment covariates.

Staggered adoption assumption: $D_{it} = 1 \Rightarrow D_{i,t+1} = 1$ for $t = 1, \ldots, T-1$. Effects may be heterogeneous across time, groups, sequences.

9.3.2.2 Parallel trends

Based on never-treated units: for all $g = 2, \ldots, T$ and $t = 2, \ldots, T$ with $t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid C = 1],$$

presuming a large and similar enough never-treated group.

Based on not-yet-treated units: for all g and s, t with $s \geq t \geq g$,

$$\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid D_s = 0, G \neq g].$$

This uses more data but restricts some pre-treatment trends (Marcus and Sant'Anna 2021). $s \geq t$ guarantees, by staggered adoption, that comparison units are untreated in $t-1$ and t.

9.3.2.3 Group-time ATT and identification

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t(g) - Y_t(0) \mid G = g],$$

e.g. $\mathrm{ATT}(2, 3)$ is the period-3 effect for units first treated in period 2. For $t \geq g$, with never-treated controls,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid C = 1];$$

with not-yet-treated controls,

$$\mathrm{ATT}(g, t) = \mathbb{E}[Y_t - Y_{g-1} \mid G = g] - \mathbb{E}[Y_t - Y_{g-1} \mid D_t = 0, G \neq g].$$

Derivation (never-treated): $\mathbb{E}[Y_t(g) \mid G = g] = \mathbb{E}[Y_t \mid G = g]$; telescoping parallel trends, $\mathbb{E}[Y_t(0) - Y_{g-1}(0) \mid G = g] = \mathbb{E}[Y_t - Y_{g-1} \mid C = 1]$; and $Y_{g-1}(0) = Y_{g-1}$ for group g (no anticipation). Each $\mathrm{ATT}(g, t)$ is a clean 2×2 DiD with base period $g - 1$.

9.3.2.4 Conditional parallel trends

Never-treated: for $t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, C = 1]$. *Not-yet-treated:* for $s \geq t \geq g \geq 2$, $\mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, G = g] = \mathbb{E}[Y_t(0) - Y_{t-1}(0) \mid X, D_s = 0, G \neq g]$. Estimation by outcome regression, IPW or DR DiD (Section 9.2).

9.3.2.5 Aggregation

By group.

$$\theta_S(g) = \frac{1}{T - g + 1} \sum_{t=2}^{T} \mathbb{1}\{g \leq t\} \, \mathrm{ATT}(g, t),$$

showing heterogeneity in adoption time, and

$$\theta_S^O := \sum_{g=2}^{T} \theta_S(g) \, P(G = g),$$

the overall effect among ever-treated groups, the recommended single summary.

By exposure length (event study).

$$\theta_D(e) := \sum_{g=2}^{T} \mathbb{1}\{g + e \leq T\} \, \mathrm{ATT}(g, g + e) \, P(G = g \mid g + e \leq T),$$

the average effect after exactly e periods of exposure.

9.3.2.6 Summary

- Group-time ATTs ATT(g,t): effect in period t for units first treated in g.
- Clean controls: not-yet-treated or never-treated units.
- Pre-treatment covariates via conditional parallel trends.
- Anticipation: shift the base period before any behavioral adjustment.

References: Athey and Imbens (2022); Baker, Larcker and Wang (2022); Callaway and Sant'Anna (2021); Chiu, Lan, Liu and Xu (2023); Goodman-Bacon (2021); Wing et al. (2024).

9.4 Synthetic control

Motivation. How can the effect on a *single* treated unit be estimated by a weighted combination of untreated units that tracks it before the intervention?

9.4.1 From DiD to synthetic controls

Motivation. Motivate a data-driven comparison unit when there is only one treated unit.

With a single treated unit, no single control may be comparable. The **synthetic control method** (SCM; Abadie and Gardeazabal 2003) uses a *convex combination* of controls whose weights minimize pre-treatment differences from the treated unit.

9.4.1.1 Case study: California Proposition 99

In 1988 California raised its cigarette tax by \$0.25 per pack (then the largest state increase). SCM builds "synthetic California" from 38 control states, mainly Utah (33.4%), Nevada (23.4%), Montana (19.9%) and Colorado (16.4%), matching pre-1988 sales, income, beer consumption, retail price and the share aged 15–24. By 2000 California's per-capita sales were about 26 packs lower than synthetic California's (Abadie, Diamond and Hainmueller 2010); the two track closely in 1970–1988, whereas the average of other states differs already before 1988. Like matching, SCM forces the researcher to show the affinity of treated and synthetic units and guards against "extreme counterfactuals" outside the convex hull.

9.4.2 The estimator

Motivation. Define the setting, estimand and weights.

9.4.2.1 Setup

Time period	1	2	$\cdots$	T_0	T_0+1	$\cdots$	T
Treated unit ($i=1$)	0	0	0	0	1	1	1
Control group ($i=2,\ldots,J+1$)	0	0	0	0	0	0	0

Unit 1 is treated from T_0+1 to T; observed $Y_{it}=Y_{it}(X_{it})$. Estimand:

$$\tau_{1t}=Y_{1t}(1)-Y_{1t}(0), \qquad t>T_0.$$

9.4.2.2 Missing counterfactuals and synthetic weights

For $t > T_0$, $\tau_{1t} = Y_{1t} - Y_{1t}(0)$, so $Y_{1t}(0)$ must be imputed. Choose weights with

$$w_j \geq 0, \qquad \sum_j w_j = 1 \quad \text{(convex)},$$

minimizing, over $t \leq T_0$,

$$\left\| Y_{1t} - \sum_{j=2}^{J+1} w_j Y_{jt} \right\| \quad \text{and} \quad \left\| Z_1 - \sum_{j=2}^{J+1} w_j Z_j \right\|,$$

with pre-treatment predictors Z_i, combined via predictor-importance weights V (Appendix 9.4.6), optionally with a dispersion penalty. Then $\hat{\tau}_{1t} = Y_{1t} - \sum_j w_j Y_{jt}$, hoping $\sum_j w_j Y_{jt} \approx Y_{1t}(0)$.

9.4.2.3 Model-based justifications

Model 1: interactive factor model.

$$Y_{it}(0) = Z_i^\top \theta_t + \alpha_i + \delta_t + \lambda_t \mu_i + \varepsilon_{it},$$

with θ_t time-varying covariate coefficients, α_i unit and δ_t time effects, λ_t a $1 \times F$ vector of common factors and μ_i an $F \times 1$ vector of loadings. $\lambda_t \mu_i$ allows structured time-varying confounding; DiD omits it.

Model 2: autoregressive model (no fixed effects):

$$Y_{i,t+1}(0) = \alpha_t Y_{it}(0) + \beta_{t+1}^\top W_{i,t+1} + u_{i,t+1}, \qquad W_{i,t+1} = \gamma_t Y_{it}(0) + \Pi_t W_{it} + v_{i,t+1},$$

with time-varying covariates W_{it}, AR coefficient α_t, feedback γ_t and AR matrix Π_t.

Properties. Suppose perfect balance: $\sum_j w_j Y_{jt} = Y_{1t}$ for $t \leq T_0$ and $\sum_j w_j Z_j = Z_1$; let $\hat{Y}_{1t}(0) = \sum_j w_j Y_{jt}$ for $t > T_0$.

- Model 1: $\hat{Y}_{1t}(0) \to Y_{1t}(0)$ as $T_0 \to \infty$ (matching many pre-period outcomes forces $\sum_j w_j \mu_j \approx \mu_1$).
- Model 2: $\mathbb{E}[\hat{Y}_{1t}(0)] = \mathbb{E}[Y_{1t}(0)]$, unbiased even with one pre-period.

9.4.2.4 Convexity: no extrapolation

Example: California's pre-treatment smoking is 120 packs per capita, donors lie in $[60, 110]$; convex weights reach at most 110, so the poor fit is visible. OLS may assign negative weights (e.g. $w_j = -0.3$) and extrapolate far outside the observed range, invisibly. SCM trades flexibility for transparency.

Sparsity: geometric intuition. The treated unit is approximated by a point on the boundary of the donors' convex hull, so only a few "vertex" donors get $w_j > 0$. Weights use no post-treatment data, protecting against specification searching.

9.4.2.5 Practical concerns

- Restrict the donor pool to similar units (less interpolation bias, overfitting).
- Donors must be unaffected by treatment (no spillovers).
- "Although we define our synthetic controls as convex combinations of unexposed units, negative weights or weights larger than one can be used at the cost of allowing extrapolation."

- Convexity may prevent a perfect pre-treatment fit outside the convex hull; a deliberate trade-off.
- With $N = 1$ treated unit, inference uses placebo tests.

9.4.3 Inference and bias

Motivation. Assess significance by placebo permutation, and understand bias from imperfect fit.

9.4.3.1 Placebo tests

Apply SCM to each of the 38 control states as if treated. Under the null, California's gap should look like a typical placebo gap; the "p-value" is the fraction of placebo gaps at least as large. In Abadie et al. (2010) only 1 of 34 states had a larger gap, $p \approx 0.03$.

In the placebo plot, gaps $Y_{jt} - \hat{Y}_{jt}^{\mathrm{SC}}$ of control states (gray) center at zero while California's (red) diverges after 1988. Exclude states with high pre-treatment RMSPE, whose gaps are uninformative.

9.4.3.2 Why imperfect pre-treatment fit causes bias

Under Model 1, a perfect pre-treatment fit gives $\sum_j w_j \mu_j \approx \mu_1$. If it fails, $\sum_j w_j \mu_j \neq \mu_1$ and

$$\sum_j w_j Y_{jt}(0) - Y_{1t}(0) \approx \lambda_t \Big(\sum_j w_j \mu_j - \mu_1 \Big) \neq 0,$$

the terms $Z_i^\top \theta_t$, α_i, δ_t cancelling by balance and ε_{it} averaging out: predictions drift after treatment, especially outside the convex hull.

9.4.3.3 Bias correction: augmented SCM

(Ben-Michael, Feller and Rothstein 2021.) Model $Y_{it}(0) = m_{it} + \varepsilon_{it}$, fit on controls ($j = 2, \ldots, J+1$), with predictions $\hat{m}_{it}$ also for $i = 1$:

$$\hat{Y}_{1t}^{\mathrm{aug}}(0) = \underbrace{\sum_{j=2}^{J+1} w_j Y_{jt}}_{\text{SC estimate}} + \underbrace{\Big(\hat{m}_{1t} - \sum_{j=2}^{J+1} w_j \hat{m}_{jt} \Big)}_{\text{bias correction}} = \underbrace{\hat{m}_{1t}}_{\text{outcome model}} + \underbrace{\sum_{j=2}^{J+1} w_j (Y_{jt} - \hat{m}_{jt})}_{\text{SC-weighted residuals}}.$$

Doubly robust: consistent if either the SC weights or the outcome model is approximately correct (cf. AIPW, DR DiD).

9.4.4 Discussion

Motivation. Weigh strengths and limitations and position SCM among alternatives.

Strengths	Limitations
Transparency: weights are explicit; one can inspect which donors contribute and how much.	Needs a good donor pool: similar units with no spillovers.
Data-driven: no need to pick a single comparison unit by hand.	Designed for a single treated unit; extensions needed for several.
No parallel-trends assumption (uses a factor model instead).	Sensitive to donor-pool and predictor choice.
Pre-treatment fit is a built-in diagnostic of credibility.	Formal inference is limited (permutation-based, not sampling-based); convexity can prevent good fit when the treated unit is an outlier.

(Abadie 2021.)

When to use SCM: few treated units, a rich pre-period, factor structure more plausible than parallel trends. Otherwise DiD, or synthetic DiD in larger panels. Open areas: conformal inference, sensitivity analysis, experimental design, external validity.

Extensions. Augmented SCM; synthetic DiD (Arkhangelsky et al. 2021: SCM unit weights plus DiD time weights and unit fixed effects); penalized SCM (Abadie and L'Hour 2021: L_1/L_2 penalties allow partial extrapolation); conformal inference (Chernozhukov, Wüthrich and Zhu 2021: permutation of residuals); matrix completion (Athey et al. 2021: low-rank imputation).

DiD versus synthetic control.

	Difference-in-differences	Synthetic control
Setup	two groups, two (or more) periods	one treated unit, many donors
Key assumption	parallel trends	pre-treatment fit (factor model)
Estimand	ATT (group level)	τ_{1t} (unit level)
Strength	simple, transparent	handles one treated unit well
Weakness	sensitive to nonlinear transforms	requires a good donor pool
Extensions	staggered DiD, nonlinear DiD	augmented SCM, matrix completion

9.4.5 Appendix: More treated units

Interactive fixed effects: estimate Model 1 directly by iterating:

(1) fix the IFE terms and fit the parametric part on untreated observations to get $\hat{\theta}$;

(2) fix the covariate coefficients and estimate the IFE terms by principal components of residuals;

(3) repeat until convergence. *Matrix completion* (Athey et al. 2021): estimate a regularized low-rank $\mathbf{L}$ approximating the observed entries of $\mathbf{Y}(0)$.

9.4.6 Appendix: Predictor importance weights V

$$w(V) = \underset{w\geq 0,\ \sum_j w_j=1}{\arg\min} \sum_{m=1}^{k} v_m \Big(X_{1m} - \sum_{j=2}^{J+1} w_j X_{jm}\Big)^2,$$

with $v_m \geq 0$ the importance of predictor m (covariate or pre-period outcome): equal $v_m = 1$; $v_m = 1/\operatorname{Var}(X_m)$; or $v_m \propto |\hat{\beta}_m|$ from an outcome regression. Abadie's standard choice is nested: $V^* = \arg\min_V \mathrm{MSPE}_{\mathrm{pre}}(w(V))$, the inner loop solving for $w(V)$.

Caveats. (1) *Outcome-lag collapse* (Kaul, Klößner, Pfeifer and Schieler 2022): with all pre-period lags, $v_m \to 0$ for other predictors; use a few lags.

(2) *Sensitivity:* V depends on predictor and lag choice; check robustness across V methods and report donor weights.

9.5 Synthetic difference-in-differences

Motivation. How can DiD's invariance to unit-level shifts and SC's re-weighting to pre-treatment trends be combined?

9.5.1 DiD, SC and SDiD

Motivation. Recall DiD and SCM and preview what SDiD changes.

Picture (Proposition 99, 1970–2000). *DiD*: California vs. the unweighted control average, all pre-periods equal; trends visibly not parallel. *SC*: re-weighted controls match California's pre-1988 level and trend; no time weights. *SDiD*: re-weighted controls parallel (not level-matched) to California, with time weights concentrated just before treatment.

Recap. DiD allows time-invariant level differences under parallel trends; SCM builds a synthetic control for one treated unit. They differ:

(1) SCM adds unit weights;

(2) SCM has time fixed effects but no unit fixed effects and no overall intercept.

SDiD (Arkhangelsky et al. 2021) re-weights to match pre-treatment trends like SCM and is invariant to additive unit shifts like DiD.

9.5.2 Formulations

Motivation. Write all three estimators as (weighted) TWFE regressions.

With unit effects α_i, time effects β_t, intercept μ and treatment X_{it},

$$\hat{\tau}^{\text{DiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2, \tag{9.1}$$

$$\hat{\tau}^{\text{SCM}} = \arg\min_{\tau,\mu,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i, \tag{9.2}$$

$$\hat{\tau}^{\text{SDiD}} = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i=1}^{N}\sum_{t=1}^{T}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\,\hat{\omega}_i\hat{\lambda}_t. \tag{9.3}$$

9.5.2.1 Setting and algorithm

Balanced panel, N units, T periods, $X_{it} \in \{0,1\}$. The first N_{co} units are never exposed; the last $N_{tr} = N - N_{co}$ are exposed after T_{pre} ($T_{post} = T - T_{pre}$). Unit weights $\hat{\omega}$ align pre-exposure trends; time weights $\hat{\lambda}$ balance pre- with post-exposure periods.

1. Compute unit weights $\hat{\omega}$.
2. Compute time weights $\hat{\lambda}$.
3. Compute $(\hat{\tau}^{\text{SDiD}}, \hat{\mu}, \hat{\alpha}, \hat{\beta}) = \arg\min_{\tau,\mu,\alpha,\beta} \sum_{i,t}(Y_{it} - \mu - \alpha_i - \beta_t - X_{it}\tau)^2\hat{\omega}_i\hat{\lambda}_t$.

9.5.2.2 Unit weights

$$(\hat{\omega}_0, \hat{\omega}) = \underset{\omega_0\in\mathbb{R},\ \omega\in\Omega}{\arg\min} \left\|\bar{y}_{pre,tr} - (\omega_0 + \omega_{co}^{\top}Y_{pre,co})\right\|_2^2 + \zeta^2 T_{pre}\|\omega_{co}\|_2^2,$$

with $\bar{y}_{pre,tr} = \frac{1}{N_{tr}}\sum_{i=N_{co}+1}^{N} Y_{it}$ the treated pre-period path, $\omega_{co}^{\top}Y_{pre,co} = \sum_{i=1}^{N_{co}}\omega_i Y_{it}$ the synthetic path, and Ω the simplex. Differences from SCM:

1. The intercept ω_0: only a *parallel trend* is needed, not the same level.
2. The L_2 penalty spreads weight over more control units.

Regularization (optional). $\zeta = (N_{tr}T_{post})^{1/4}\,\hat{\sigma}(\Delta_{it})$, with $\Delta_{it} = Y_{i(t+1)} - Y_{it}$ the pre-period first differences of controls,

$$\bar{\Delta} = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}\Delta_{it}, \qquad \hat{\sigma}^2(\Delta_{it}) = \frac{1}{N_{co}(T_{pre}-1)}\sum_{i=1}^{N_{co}}\sum_{t=1}^{T_{pre}-1}(\Delta_{it} - \bar{\Delta})^2.$$

9.5.2.3 Time weights

$$(\hat{\lambda}_0, \hat{\lambda}) = \underset{\lambda_0 \in \mathbb{R},\ \lambda \in \Lambda}{\arg\min} \left\| \bar{y}_{post,co} - (\lambda_0 + \lambda_{pre}^\top Y_{pre,co}) \right\|_2^2,$$

with $\bar{y}_{post,co} = \frac{1}{T_{post}} \sum_{t=T_{pre}+1}^{T} Y_{it}$ the $1 \times N_{co}$ post-period control averages, $Y_{pre,co}$ the $T_{pre} \times N_{co}$ pre-period control outcomes, $\lambda_{pre} \in \Lambda$ (simplex) a $1 \times T_{pre}$ vector, and λ_0 an intercept letting the post-period lie above or below all pre-periods.

9.5.3 Comparisons and takeaways

Motivation. Explain what the weights and fixed effects buy.

SDiD versus DiD. DiD has no weights. The weights make the regression "local", emphasizing units and periods similar to the treated ones: more robust, and more precise by removing predictable parts of the outcome.

SDiD versus SCM. SCM omits unit fixed effects and time weights:

$$\hat{\tau}^{\text{SCM}} = \underset{\mu,\beta,\tau}{\arg\min} \sum_{i,t} (Y_{it} - \mu - \beta_t - X_{it}\tau)^2 \hat{\omega}_i^{\text{SCM}}.$$

Time weights drop periods unlike the post-period (less bias, more precision); unit fixed effects add flexibility and explain much outcome variation.

Key takeaways.

- Like SCM, SDiD works when pre-treatment trends are not parallel.
- Unlike SCM, its weights ω give a control only *parallel* to the treated group.
- Like DiD, it uses time (β) and unit (α) fixed effects, reducing variance.
- Its unit weights use an L_2 penalty and an intercept ω_0.
- It adds time weights, which neither DiD nor SCM has.